EL-MOASSER

By A Group Of Supervisors

PHYSICS
PART 2

QUESTIONS



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CONTENTS

UNIT TWO : Introduction to Modern Physics

Chapter 5

Wave Particle Duality



Lesson One
- Blackbody Radiation.
- Thermionic Effect and Photoelectric Effect.

Lesson Two
- Compton Effect.
- Wave Particle Duality.
- Electron Microscope.

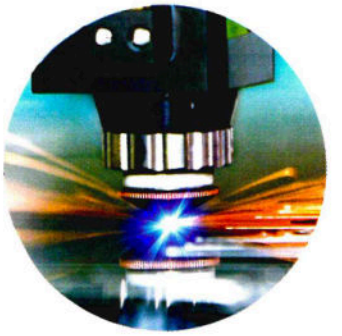
Chapter 6

Atomic Spectra



Chapter 7

Laser



Chapter 8

Modern Electronics



Lesson One
- The Semiconductor Crystal.
- The pn Junction.

Lesson Two
- The Transistor.
- Digital and Analog Electronics.

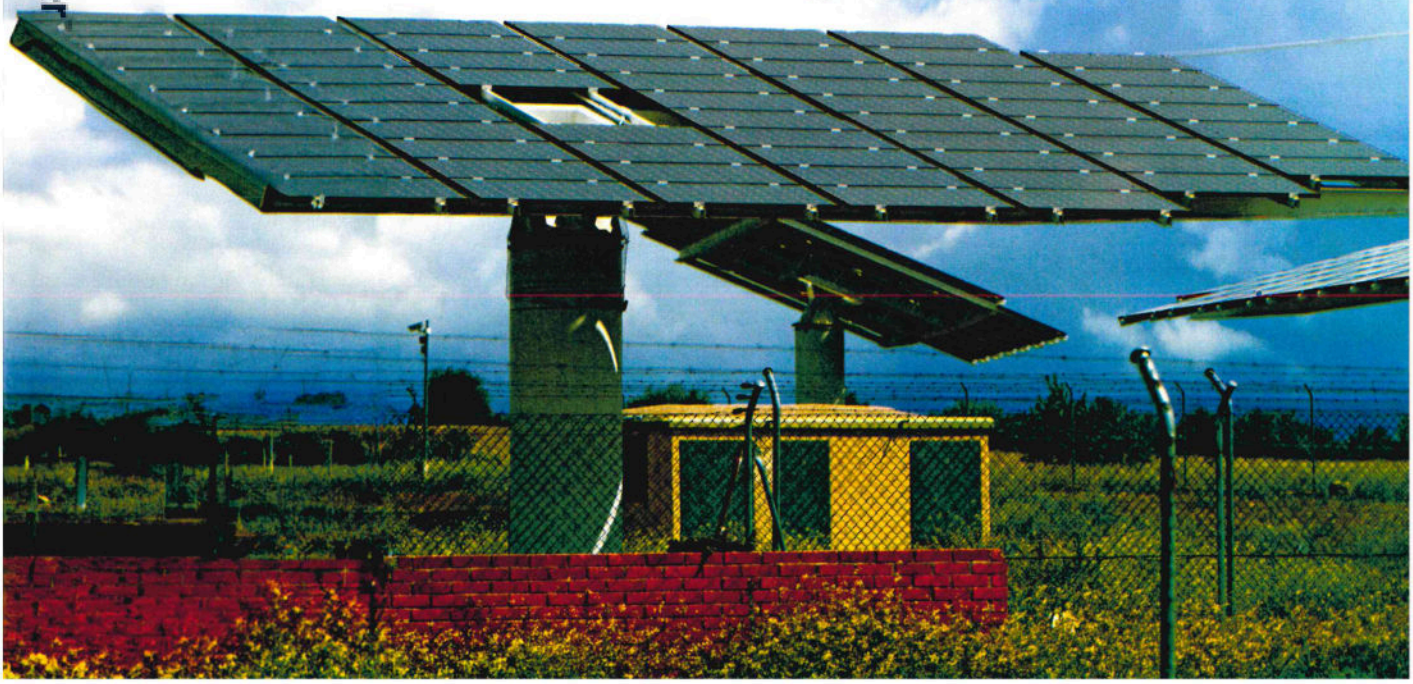
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Unit Two

Introduction to Modern Physics

Chapter Five	Wave Particle Duality.
Chapter Six	Atomic Spectra.
Chapter Seven	Laser.
Chapter Eight	Modern Electronics.

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CHAPTER

FIVE

Wave Particle Duality

Lesson 1
- Blackbody Radiation.
- Thermionic Effect and Photoelectric Effect.

Lesson 2
- Compton Effect.
- Wave Particle Duality.
- Electron Microscope.

? • Questions on each lesson from the Previous Years Exams.



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Chapter 5

Questions on Lesson One

• Blackbody Radiation.
• Thermionic Effect and Photoelectric Effect.

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

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Important guidelines

• Wien's law:

$$\frac{(\lambda_{max})_1}{(\lambda_{max})_2} = \frac{T_2}{T_1}$$

The emission of the electrons from a metallic surface

Photoelectric effect

$E = E_w + (KE)_{max}$

$E = h\upsilon = \frac{hc}{\lambda}$

$E_w = h\upsilon_c = \frac{hc}{\lambda_c}$

$(KE)_{max} = \frac{1}{2} m_e v^2$

Cathode ray tube

$(KE)_{max} = \frac{1}{2} m_e v^2 = eV$

• The energy (in joule) = The energy (in electron volt) × Charge of the electron

First Multiple choice questions

Use the following constants if needed:

$c = 3 \times 10^8$ m/s, $h = 6.625 \times 10^{-34}$ J.s, $m_e = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C

Blackbody radiation

1. In the electromagnetic spectrum, the ratio between the wavelength of the red light rays and the wavelength of the ultraviolet rays is ..... .

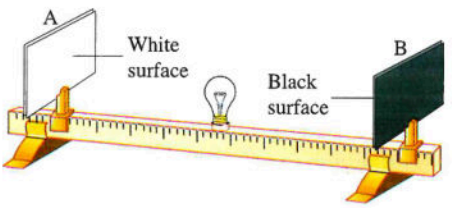
   (a) greater than one

   (b) less than one

   (c) equal to one

   (d) equal to the ratio between the speeds of the two rays

2. In the opposite figure, when the bulb is lit during a certain period, **which** of the two surfaces A or B its temperature increases more?



   (a) Surface A.

   (b) Surface B.

   (c) The two surfaces have equal temperature.

   (d) The answer is indeterminable.

3. The dominant color of the emitted electromagnetic radiation from an ideal blackbody changes according to ..... .

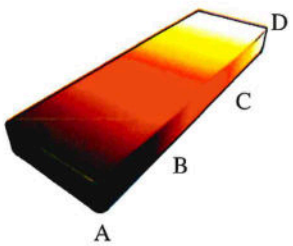
   (a) the mass of the body

   (b) the density of the material of the body

   (c) the temperature of the body

   (d) the type of the material of the body

4. The opposite figure shows a piece of heated iron.
   Which position on the iron piece has the least temperature?



   (a) A

   (b) B

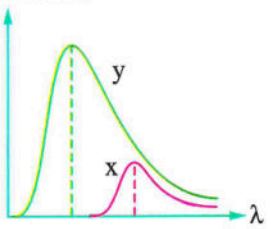
   (c) C

   (d) D

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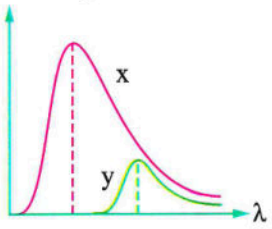
5 Which of the following graphs represents Planck's curve of a radiation emitted from two identical bodies x, y whose temperatures are constant at T_x, T_y respectively, where $(T_x > T_y)$?

Radiation intensity



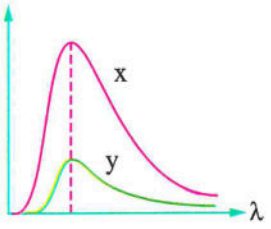
(a)

Radiation intensity



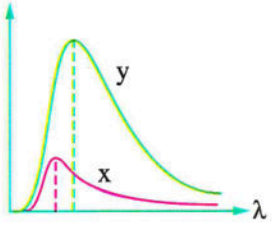
(b)

Radiation intensity



(c)

Radiation intensity



(d)

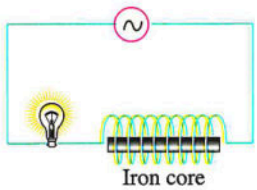
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6 In the blackbody radiation curve, if the temperature of a body gets decreased, ...

	The peak of the curve	The area under the curve
(a)	displaces to the high frequencies region	increases
(b)	displaces to the high frequencies region	decreases
(c)	displaces to the low frequencies region	increases
(d)	displaces to the low frequencies region	decreases

---

7 An AC circuit consists of an AC power source connected to a lamp and an induction coil containing an iron core, as shown in the opposite figure, so the peak wavelength (λ_o) of the radiation emitted by the lamp has a maximum intensity (I_o).



If the iron core is removed from the inductive coil then ..... .

	The maximum intensity of radiation	The wavelength corresponding to maximum intensity
(a)	remains I_o	remains λ_o
(b)	remains I_o	decreases below λ_o
(c)	increases above I_o	remains λ_o
(d)	increases above I_o	decreases below λ_o

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8 Two identical bodies x and y, if the temperature of body x is greater than the temperature of body y, the ratio between the total radiation energy of body x and that of body y is ..... .

(a) less than one

(b) equal to one

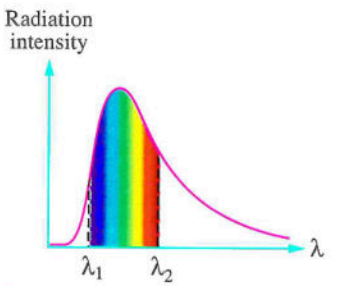
(c) greater than one

(d) the information is not enough to determine the answer

9 The intensity of the radiation at high frequencies in Planck's curve for the radiation of a blackbody ..... .

(a) doesn't change (b) is moderate (c) is very large (d) approaches zero

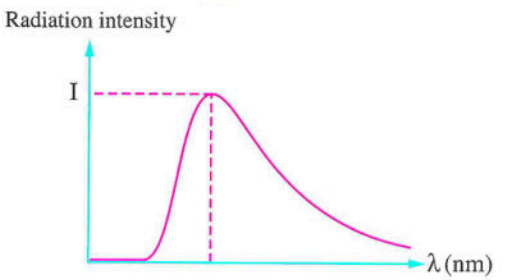
10 In the opposite graph, if λ_1 is the shortest wavelength of the visible light and λ_2 is the longest wavelength of the visible light, the graph may represent the radiation of ..... .



(a) a luminous star (b) the Earth

(c) a tungsten bulb (d) the human body

11 The opposite graph represents Planck's curve for a hot blackbody, what happens to each of the following when the temperature increases:



(i) The number of photons emitted from the body in one second?

(a) Increases. (b) Decreases.

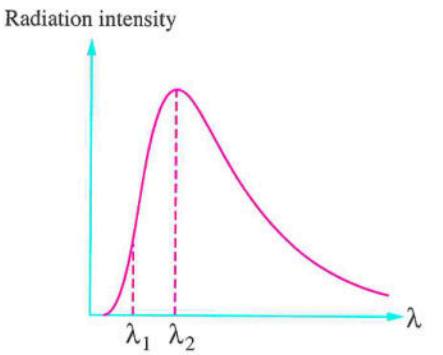
(c) Doesn't change. (d) Is indeterminable.

(ii) The value of the maximum radiation intensity?

(a) Becomes greater than I. (b) Becomes less than I.

(c) Remains equal to I. (d) Is indeterminable.

12 The opposite graph represents Planck's curve for a hot ideal blackbody radiation, the rate of photons emission with wavelength (λ_1) compared to the rate of their emission with wavelength (λ_2) is ..... .



(a) greater

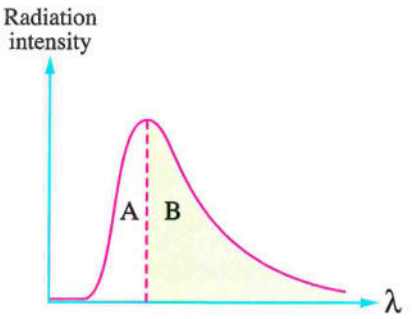
(b) smaller

(c) equal

(d) indeterminable

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13 The opposite graph represents Planck's curve of the radiation of an ideal blackbody, which of the following two regions disobeys the classical physics principles?



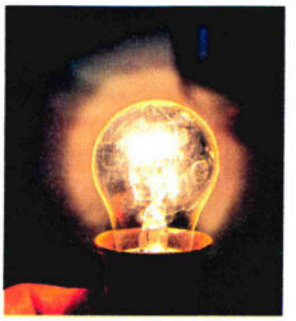
(a) Region A.

(b) Region B.

(c) Both regions A and B.

(d) No correct answer.

14 The opposite figure shows an incandescent electric bulb, so the ratio between the total energy of the produced invisible radiation and the total energy of the produced visible light through the same period of time is ..... .



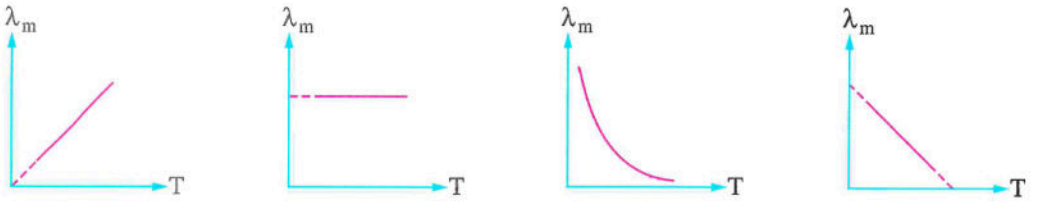
(a) greater than one

(b) less than one

(c) equal to one

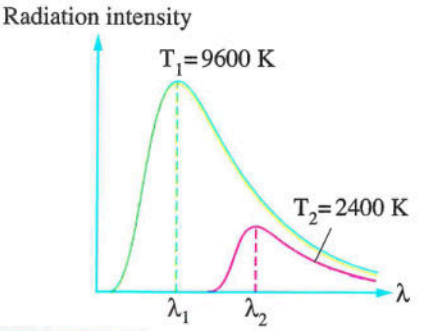
(d) not changed by the change of temperature

15 Which of the following graphs represents the relation between the wavelength (λ_m) associated with the maximum intensity of radiation emitted from many similar hot bodies and absolute temperature (T) for each of them?



(a) (b) (c) (d)

16 The opposite graph represents Planck's distribution for a hot blackbody at two temperatures T_1 and T_2, then the ratio $\left(\frac{\lambda_1}{\lambda_2}\right)$ is ..... .



(a) $\frac{1}{16}$ (b) $\frac{1}{8}$

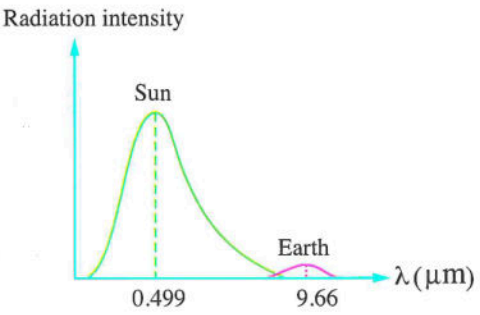
(c) $\frac{1}{4}$ (d) $\frac{1}{2}$

17 A blackbody of temperature 3000 K and the wavelength which has radiation of maximum intensity is λ. When it is cooled to an absolute temperature T, the wavelength which has radiation of maximum intensity becomes 10 λ, then the temperature T is ..... .

(a) 300 K (b) 2700 K (c) 270 K (d) 1800 K

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18 ✳ The opposite figure represents the relation between the intensity of the emitted radiation from two hot bodies and the wavelength (λ). If you know that the temperature of the Sun's surface is 6000 K, then using the given data in the figure, the average temperature of the Earth's surface is ..... .

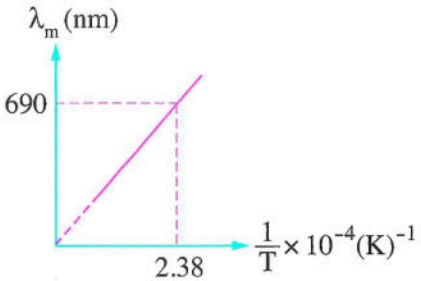


(a) 9000 K (b) 1935.9 K
(c) 309.9 K (d) 200 K

19 A blackbody, the wavelength associated with the maximum radiation intensity of the emitted radiation at absolute temperature (T_o) is (λ_o), if the temperature of body increases by 500 K, the wavelength associated with the maximum radiation intensity emitted from the body decreases by ($\frac{5}{8}\lambda_o$), therefore the temperature (T_o) equals ..... .

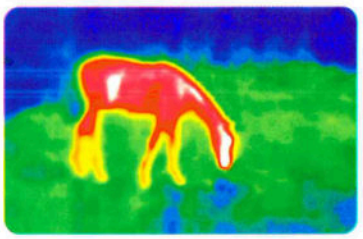
(a) 300 K (b) 500 K (c) 800 K (d) 1500 K

20 The opposite graph represents the relation between the wavelength (λ_m) associated with the maximum radiation intensity emitted from many hot bodies and the reciprocal of absolute temperature ($\frac{1}{T}$) for each of them, so the wavelength associated with the maximum radiation intensity emitted from a body of temperature 4830 K is found in the region of ..... .



(a) infrared rays (b) visible light (c) ultraviolet rays (d) X-rays

21 The opposite figure shows a photo taken by one of the night vision devices, so the device depends on receiving what the bodies emit from ..... .



(a) visible radiation (b) UV radiation
(c) infrared rays (d) X-rays

22 According to quantum mechanics assumptions of Max Planck:

(i) The atoms of the blackbody emit photons ..... .

(a) when transferring from ground level to high energy level
(b) have different wavelengths but the same intensity
(c) whose number decreases when the frequency of radiation increases with a certain limit
(d) whose energy is directly proportional to the wavelength of radiation

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(ii) When an atom transfers from a high energy level to a low energy level, it ..... .

(a) absorbs a photon of energy $h\upsilon$ (b) absorbs a photon of energy $h\upsilon^2$

(c) emits a photon of energy $h\upsilon$ (d) emits a photon of energy $h\upsilon^2$

(iii) The energy of a photon whose frequency is 5×10^{14} Hz equals ..... .

(a) 1.7×10^{-19} J (b) 3.3×10^{-19} J (c) 6.6×10^{-19} J (d) 9.9×10^{-19} J

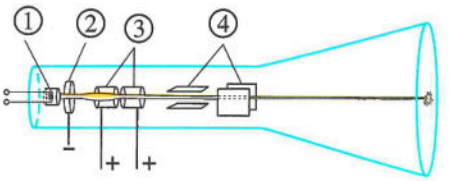
Thermionic effect and photoelectric effect

23 One of the characteristics of the cathode rays that they ..... .

(a) are electromagnetic waves (b) have a constant velocity

(c) are charged particles (d) are uncharged particles

24 The opposite figure represents a cathode ray tube:



(i) Which part in the tube is the source of electrons?

(a) Part ① (b) Part ②

(c) Part ③ (d) Part ④

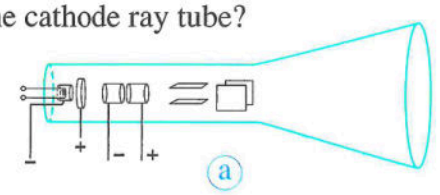
(ii) Which part of the tube is responsible for changing the position of the electron beam on the screen?

(a) Part ① (b) Part ② (c) Part ③ (d) Part ④

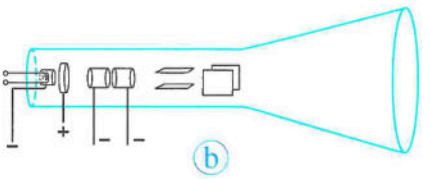
(iii) Which part in the tube is responsible for controlling the light intensity at the fluorescent screen?

(a) Part ① (b) Part ② (c) Part ③ (d) Part ④

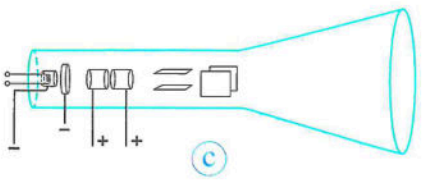
25 Which figure shows the correct electric potentials used to form an image on the screen of the cathode ray tube?



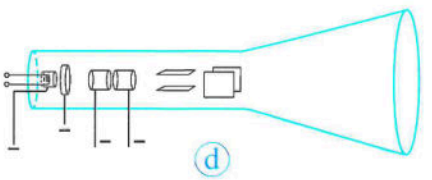
(a)



(b)



(c)



(d)

26 In the CRT, when the grid potential changes from –20 V to –50 V, ...

(a) the illumination at the fluorescent screen increases

(b) the illumination at the fluorescent screen decreases

(c) the deflection of the electron beam increases

(d) the deflection of the electron beam decreases

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27 In the CRT, when the filament gets burnt, ...

(a) the illumination at the fluorescent screen increases

(b) the illumination at the fluorescent screen decreases

(c) the fluorescent screen does not emit light

(d) the deflection of the electron beam decreases

28 In the cathode ray tube, when the grid is not connected to any electrical signal, ...

(a) the path of the electron beam to the screen cannot be controlled

(b) the fluorescent screen does not emit light

(c) the electron beam returns to the cathode

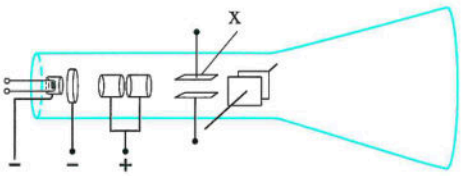
(d) the brightness of the screen remains almost constant

29 Which of the following choices represents what appears on the screen of the cathode ray tube when the deflection system of the beam stops functioning?



(a) (b) (c) (d)

30 What happens to the path of the cathode ray when a DC voltage is connected between the terminals of part (x)?



(a) The electron beam deflects towards the plate of negative potential.

(b) The electron beam deflects towards the plate of positive potential.

(c) The electron beam does not deflect.

(d) The electron beam does not pass through part (x).

31 The kinetic energy of the electrons when reaching the anode in the cathode ray tube depends on ..... .

(a) the surface area of the cathode

(b) the type of the anode material

(c) the potential difference between the plates of the beam deflection system

(d) the potential difference between the anode and the cathode

32 In cathode ray tube, if the potential difference between the anode and cathode increases and the temperature of the filament doesn't change, the rate of electron emission from the filament ..... .

(a) increases (b) decreases

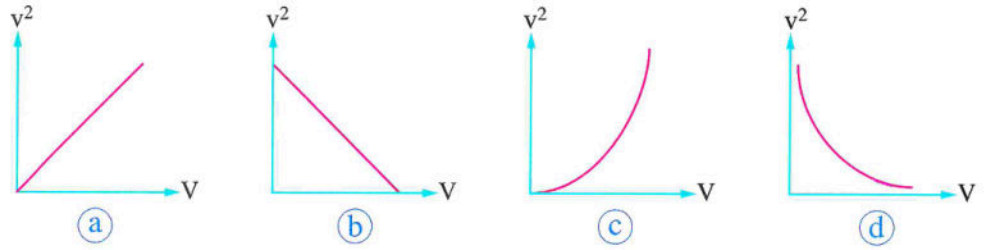
(c) doesn't change (d) cannot be determined

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33 In CRT, when an electron is accelerated by a potential difference equals (V), it moves with a maximum speed (v). If the potential difference acting on the electron is increased to 2 V, the maximum speed of the electron becomes ..... .

(a) 2 v (b) $\sqrt{2}$ v (c) 4 v (d) $\frac{1}{2}$ v

34 The graph that represents the relation between the square of the maximum speed (v^2) of the emitted electrons that reach the anode in the cathode ray tube and the potential difference (V) between the anode and cathode in the cathode ray tube is ..... .



(a) (b) (c) (d)

35 ✳ If the potential difference between the cathode and the anode in a cathode ray tube is 1000 V, then:

(i) The maximum kinetic energy of the electrons is ..... .

(a) 1.6×10^{-15} J (b) 1.6×10^{-16} J (c) 1.6×10^{-17} J (d) 32×10^{-18} J

(ii) The maximum speed of the emitted electron from the cathode when it reaches the anode equals ..... .

(a) 1.88×10^{6} m/s (b) 1.88×10^{7} m/s (c) 3.52×10^{5} m/s (d) 3.52×10^{4} m/s

36 The photoelectric emission from the cathode's surface of the photoelectric cell depends on ..... .

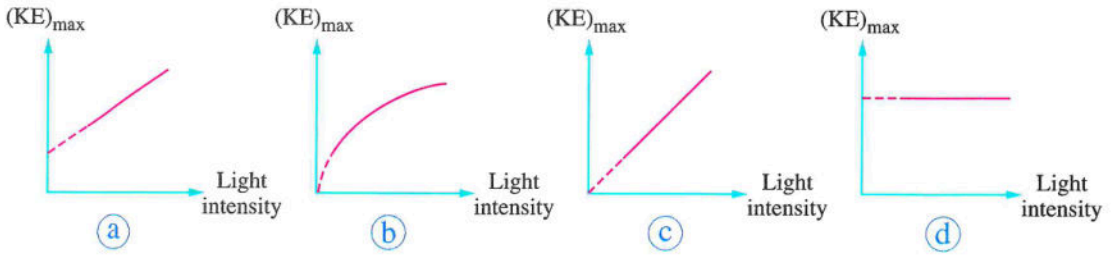
(a) the type of the anode material and the intensity of the incident light

(b) the type of the cathode material and the frequency of the incident light

(c) the type of the anode material and the frequency of the incident light

(d) the type of the cathode material and the intensity of the incident light

37 Which of the following graphs represents the relation between the maximum kinetic energy $(KE)_{max}$ for the emitted electrons from the cathode of a photoelectric cell and the intensity of the monochromatic incident light on the cathode?



(a) (b) (c) (d)

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38 In the photoelectric cell, when an electromagnetic radiation of a certain frequency falls on the cathode of the cell, electrons are emitted from it with a certain maximum kinetic energy. If the incident radiation on the cathode is replaced with another of higher frequency, the quantity that remains unchanged is ..... .

(a) the energy of the incident photon
(b) the speed of the incident photon
(c) the maximum energy of the emitted electrons
(d) the maximum speed of the emitted electrons

39 In the photoelectric effect, the ratio between the energy of the incident photon (E_1) on a metal surface and the kinetic energy of the liberated electron (E_2) from the surface $\left(\frac{E_1}{E_2}\right)$ is ..... .

(a) greater than one (b) less than one (c) equal to one (d) indeterminable

40 In an experiment for the photoelectric cell, the surface of the cathode is exposed to a monochromatic light of frequency greater than the critical frequency of the metal, then the experiment is repeated using a light of the same wavelength but its intensity is double that of the first, therefore the quantity that is doubled is ..... .

(a) the energy of the photon
(b) the maximum kinetic energy of the emitted electrons
(c) the work function of the metal
(d) the intensity of the photoelectric current

41 When a light ray of wavelength 591.5 nm has fallen on the surface of the cesium metal, photoelectrons are barely emitted from its surface, so to emit electrons of kinetic energy 1.81 eV, the wavelength of the incident light ray should be equal to ..... approximately.

(a) 318 nm (b) 520 nm (c) 650 nm (d) 720 nm

42 When an electromagnetic ray of a single wavelength falls on the cathode of a photoelectric cell, a photoelectric current is detected in its circuit, the emission rate of electrons from the cathode of a photoelectric cell decreases but doesn't vanish by decreasing ..... .

(a) the wavelength of the incident light (b) the frequency of the incident light
(c) the speed of the incident light (d) the intensity of the incident light

43 Which of the following factors controls the rate of electron emission from a metallic surface when the surface gets exposed to incident photons whose wavelength is less than the critical wavelength of the metal surface?

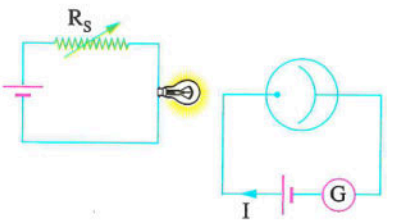
(a) The photons' beam frequency. (b) The photon's energy.
(c) The intensity of photons' beam. (d) The linear momentum of the photon.

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44 When a monochromatic light has fallen on the surface of a certain metal, a number of electrons is emitted, so if another monochromatic light of higher frequency has fallen on the surface of the same metal with the same rate of incident photons as the first light, the number of emitted electrons per second ..... .

(a) increases (b) decreases (c) remains unchanged (d) is indeterminable

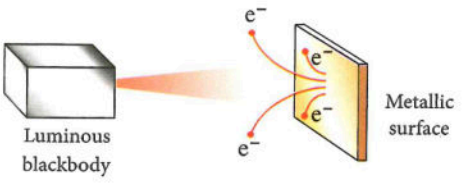
45 In the opposite figure, light is emitted from an electric bulb and falls on a photoelectric cell causing the flow of a photocurrent, if the brightness of the bulb increases, the photocurrent intensity ..... .



(a) increases (b) decreases but doesn't vanish

(c) doesn't change (d) vanishes

46 In the opposite figure, a luminous blackbody is placed in front of a metal surface so that the emitted radiation from the blackbody has caused electrons emission from the metal surface. If the temperature of the blackbody is raised, then:



(i) The rate of electron emission from the metal surface ..... .

(a) increases (b) decreases (c) doesn't change (d) vanishes

(ii) The maximum kinetic energy of the emitted electrons from the metal surface ..... .

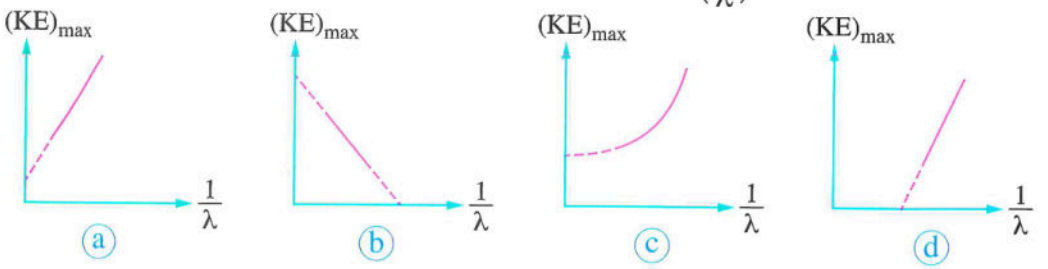
(a) increases (b) decreases (c) doesn't change (d) vanishes

47 If the frequency of the incident light on a metallic surface is doubled, the work function of this metal ..... .

(a) increases to the double (b) decreases to the half

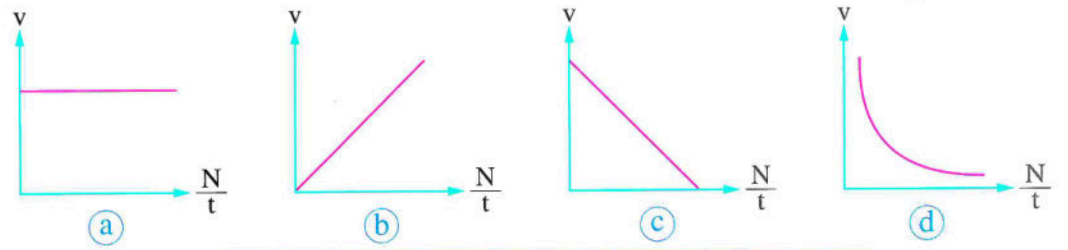
(c) remains constant (d) quadruples

48 Which of the following graphs represents the relation between the maximum kinetic energy $(KE)_{max}$ of the photoelectrons and the reciprocal of the wavelength of the incident rays on the cathode of the photoelectric cell $\left(\frac{1}{\lambda}\right)$?

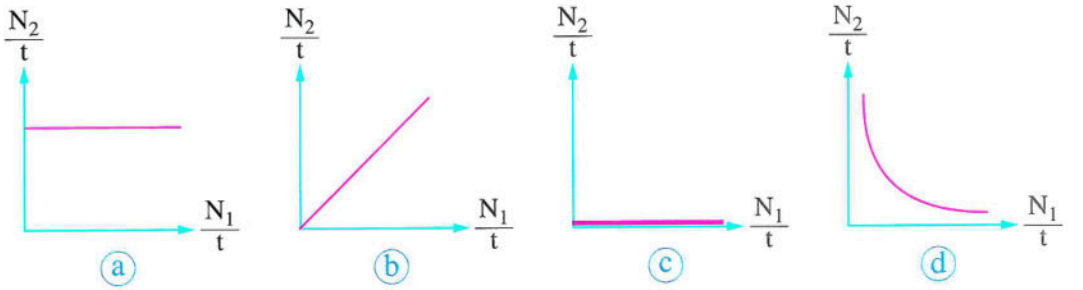


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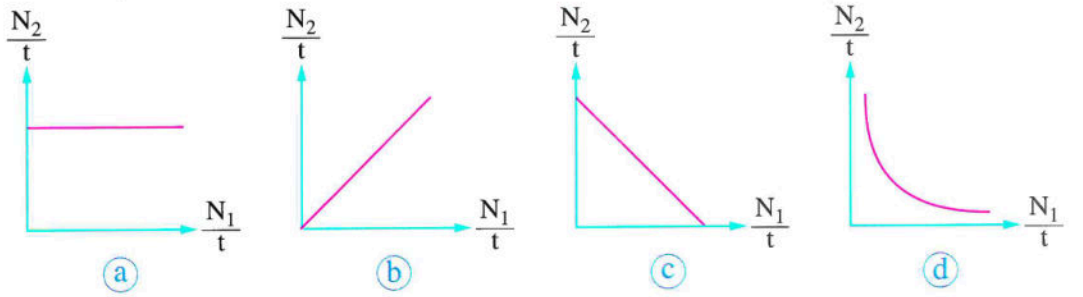
49 A light has fallen on a metal surface such that its frequency was greater than the critical frequency of that surface. Which of the following graphs represents the relation between the speed of the emitted electrons (v) and the incidence rate of photons $\left(\frac{N}{t}\right)$ on the surface?



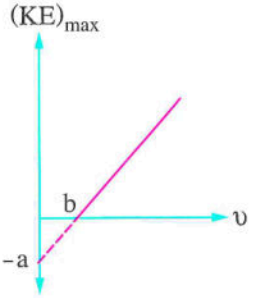
50 A light has fallen on a metal surface such that its frequency was less than the critical frequency of that surface. Which of the following graphs represents the relation between the emission rate of electrons $\left(\frac{N_2}{t}\right)$ from that surface and the rate of incident photons $\left(\frac{N_1}{t}\right)$ on the surface?



51 When a light has fallen on a metal surface such that its frequency was greater than the critical frequency of that surface, which of the following graphs represents the relation between the emission rate of electrons $\left(\frac{N_2}{t}\right)$ from that surface and the rate of the incident photons $\left(\frac{N_1}{t}\right)$ on the surface?



52 The opposite graph represents the relation between the maximum kinetic energy $(KE)_{max}$ of the electrons emitted from the cathode of a photoelectric cell and the frequency (υ) of the falling light on the cathode of that cell, then the ratio between (a) and (b) represents ..... .

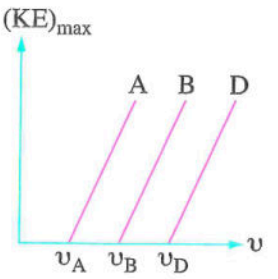


(a) Planck's constant (b) critical frequency

(c) work function (d) photon energy

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53 The opposite graph represents the relation between the frequency of the incident light rays on the surfaces of three metals A, B and D and the maximum kinetic energy of the emitted electrons, if the work functions for these metals are E_A, E_B and E_D, then, ...



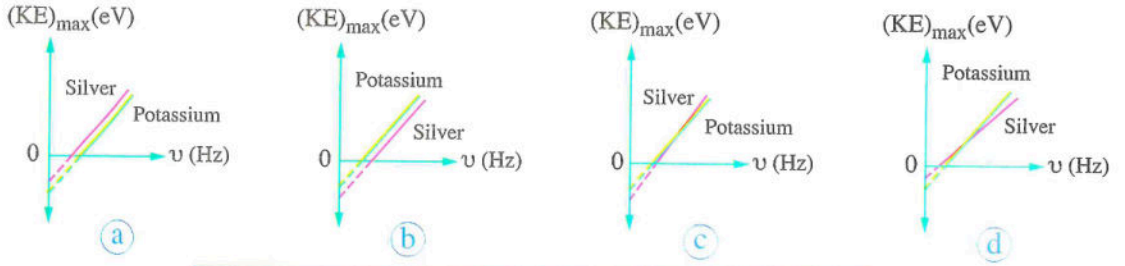
(a) $E_A < E_B < E_D$ (b) $E_A = E_B < E_D$

(c) $E_A > E_B > E_D$ (d) $E_A = E_B = E_D$

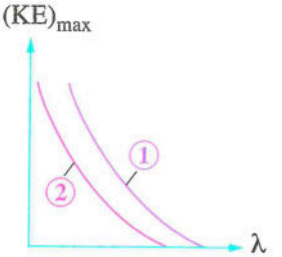
54 Three metallic surfaces (x), (y) and (z), the two surfaces (x), (y) are from cesium and surface (z) is from silver. Electromagnetic radiations of wavelengths $\lambda, \frac{2}{3}\lambda, \lambda$ fall on each of them, respectively, so photoelectric electrons are emitted from the three surfaces, if you know that the work function of cesium is less than that of silver, then the correct order for the three surfaces according to the maximum velocity of the emitted electrons is ..... .

(a) x > y > z (b) x < y < z (c) y > z = x (d) y > x > z

55 If you know that the work function for each of potassium and silver are 1.8 eV and 4.73 eV respectively, which of the following graphs represents the relation between the maximum kinetic energy (KE_{max}) of the emitted electrons from each of them and the frequency of the incident light (υ) on each of them?



56 The opposite graph represents the variation between the maximum kinetic energy $(KE)_{max}$ for the emitted electrons from two metallic surfaces (1) and (2) versus the wavelength (λ) of the incident photons on each of them, Which of the two metals has the greatest work function?

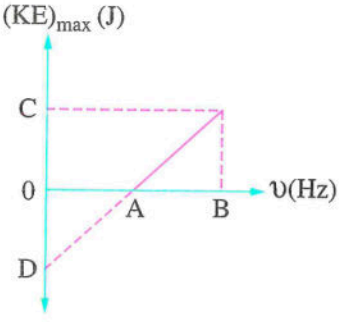


(a) Metal (1). (b) Metal (2).

(c) The two metals have the same work function.

(d) The answer cannot be determined.

57 ✳ The opposite graph represents the relation between the maximum kinetic energy of the emitted electrons from the surface of the cathode of a photoelectric cell and the frequency of the incident light on it, then which of the following quantities represents Planck's constant?

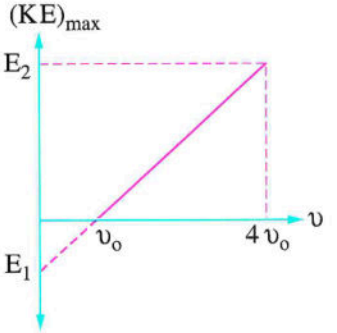


(a) D (b) $\frac{D}{B+A}$

(c) $\frac{A}{B}$ (d) $\frac{C}{B-A}$

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58 The opposite graph represents the relation between the maximum kinetic energy $(KE)_{max}$ of the emitted electrons from a metallic surface and the frequency (υ) of the incident light, so the ratio $\left(\frac{E_1}{E_2}\right)$ equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{1}{4}$ (d) $\frac{1}{5}$

59 An electromagnetic radiation of a single wavelength falls on a metallic surface of work function 3 eV, which of the following can represent the photon's energy incident on this surface and the maximum value of kinetic energy of the emitted electrons?

	The incident photon's energy (eV)	The maximum value of kinetic energy of the electrons (eV)
(a)	2	1
(b)	2	5
(c)	4	1
(d)	4	2

60 An electromagnetic radiation of frequency υ has fallen on the surface of a metal of work function 3 eV, so electrons of maximum kinetic energy 2 eV are emitted from its surface. If the incident radiation is replaced by another of frequency 2 υ, the maximum kinetic energy of the emitted electrons equals ..... .

(a) 7 eV (b) 6 eV (c) 5 eV (d) 4 eV

61 A light ray of frequency 5 υ_c has fallen on a metal surface of critical frequency υ_c, so photoelectrons are emitted with a maximum speed of 8×10^6 m/s. If the frequency of the falling ray became 2 υ_c, the maximum speed of the emitted electrons becomes ..... .

(a) 6×10^6 m/s (b) 4×10^6 m/s (c) 8×10^6 m/s (d) 1×10^6 m/s

62 An electromagnetic radiation of a single wavelength falls on two metallic surfaces (x), (y) whose work functions are 3 eV and 4 eV, respectively, so the maximum value of the kinetic energy of the electrons emitted from the two surfaces, respectively, may equal ..... .

(a) 1 eV, 2 eV (b) 2 eV, 1 eV (c) 3 eV, 1 eV (d) 2 eV, 2 eV

63 In a photoelectric cell experiment, when an electromagnetic radiation of wavelength λ was used, the maximum kinetic energy for the emitted electrons was KE. If another radiation of wavelength $\frac{\lambda}{2}$ is used, the maximum kinetic energy for the emitted electrons will be ..... .

(a) equal to zero
(b) less than 2 KE and greater than KE
(c) 2 KE
(d) greater than 2 KE

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64 ✳ When a light of wavelength 623 nm has fallen on the surface of a metal, electrons are liberated with a speed of 4.6×10^5 m/s, then:

(i) The work function of that surface is ..... .

(a) 2.23×10^{-19} J (b) 3.19×10^{-19} J (c) 4.15×10^{-19} J (d) 4.6×10^{-19} J

(ii) The critical frequency of that surface is ..... .

(a) 6.94×10^{14} Hz (b) 6.26×10^{14} Hz (c) 4.81×10^{14} Hz (d) 3.37×10^{14} Hz

65 In a photoelectric cell, when yellow light has fallen on the surface of the cathode, no electrons are emitted from the surface but when blue light has fallen on the surface of the same cathode, electrons are emitted at a certain rate. So, if red light falls on the surface of the same cathode, the emission rate of the electrons ..... .

(a) increases (b) decreases but does not vanish

(c) vanishes (d) doesn't change

66 If the critical frequency of a metal surface lies in the blue light region, the incident rays that allow the emission of electrons from the surface of that metal are ..... .

(a) IR rays (b) UV rays (c) radio waves (d) red rays

67 Four photons x, y, z and k whose energies are 6 eV, 5 eV, 4 eV, and 3 eV respectively, have fallen one at a time on a metallic surface with a work function E_w, so three electrons are emitted from that surface then the work function E_w of this surface is ..... .

(a) $6\text{ eV} > E_w > 5\text{ eV}$ (b) $5\text{ eV} > E_w > 4\text{ eV}$

(c) $4\text{ eV} > E_w > 3\text{ eV}$ (d) $3\text{ eV} > E_w$

68 An electromagnetic radiation of wavelength 3100 Å has fallen on the cathode of a photoelectric cell, so photoelectrons are emitted whose maximum kinetic energy was 2.5 eV, then the work function of the cathode material is ..... .

(a) 3.1 eV (b) 2.4 eV (c) 1.5 eV (d) 0.9 eV

69 If the work function of a metallic surface is 3.3125×10^{-19} J, the critical frequency of this metal equals ..... .

(a) 4.5×10^{14} Hz (b) 4.8×10^{14} Hz (c) 5×10^{14} Hz (d) 5.5×10^{14} Hz

70 If the work function of a metallic surface is 2.48 eV, the longest wavelength of incident light to liberate electrons from the surface equals ..... .

(a) 4×10^{-7} m (b) 5×10^{-7} m (c) 5.5×10^{-7} m (d) 6×10^{-7} m

71 ✳ When a monochromatic light (a) of wavelength 4000 Å has fallen on a metallic surface, electrons are emitted with a speed that equals 5.3×10^5 m/s. If another monochromatic light (b) of wavelength 5500 Å has fallen on the same surface, then the light (b) ..... .

(a) has frequency greater than the critical frequency of the metal

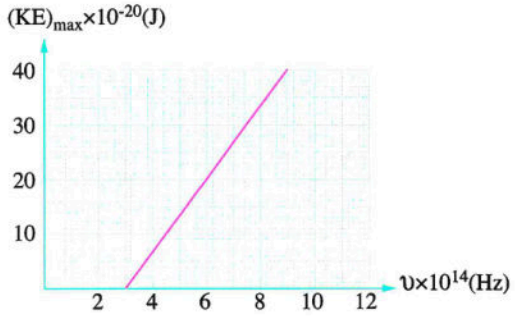
(b) has frequency less than the critical frequency of the metal

(c) has frequency equals the critical frequency of the metal

(d) liberates electrons from the metal with higher speed

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72 ✳ The opposite graph represents the relation between the maximum kinetic energy $(KE)_{max}$ of the emitted electrons from a metallic surface and the frequency (υ) of incident light, from the graph:



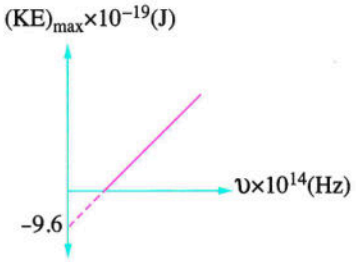
(i) The critical frequency of the metal surface equals ..... .

(a) 2.4×10^{14} Hz (b) 2.5×10^{14} Hz (c) 3×10^{14} Hz (d) 4.8×10^{14} Hz

(ii) The wavelength of light which causes the emission of electrons with a maximum kinetic energy 20×10^{-20} J equals ..... .

(a) 1×10^{-6} m (b) 5×10^{-7} m (c) 1.67×10^{-15} m (d) 3×10^{-15} m

73 ✳ Different monochromatic light rays have fallen on the cathode of a photoelectric cell and the opposite graph represents the relation between the maximum kinetic energy $(KE)_{max}$ of the emitted electrons and the frequency (υ) of the incident light, then:



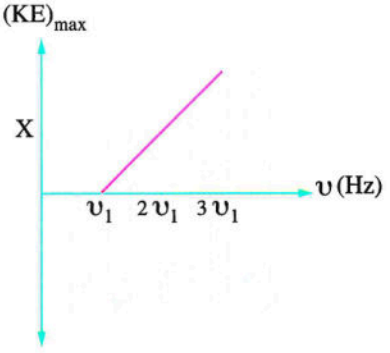
(i) The critical wavelength for the cathode material is ..... .

(a) 2.07×10^{-7} m (b) 6.9×10^{-7} m (c) 2.07×10^{-8} m (d) 6.9×10^{-8} m

(ii) The frequency of the incident light that liberates an electron of kinetic energy 9.6×10^{-19} J from the cathode surface is ..... .

(a) 1.45×10^{15} Hz (b) 2.9×10^{15} Hz (c) 5.8×10^{15} Hz (d) 19.2×10^{15} Hz

74 The opposite figure represents the relation between the maximum kinetic energy $(KE)_{max}$ of the emitted electrons from a metal surface and the frequency (υ) of the falling light on the surface of that metal. If the work function for that metal is 5×10^{-19} J, the value of X is equal to ..... .



(a) 5×10^{-19} J (b) 1×10^{-19} J
(c) 2×10^{-19} J (d) 3×10^{-19} J

75 A photon of energy 3 hυ has fallen on a metallic surface of work function hυ, then this photon can liberate a number of electrons that equals ..... .

(a) zero (b) 1 (c) 2 (d) 3

76 A light ray of frequency υ falls on a metal surface of work function E_w, so the maximum kinetic energy of the emitted electrons was KE. If the frequency of the incident light is doubled, the maximum kinetic energy of the emitted electrons becomes .....

(a) 2 KE (b) 2 KE – E_w (c) 2 KE + E_w (d) 2 KE + 2 E_w

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77 ✳ If the critical wavelength of zinc is 3000 Å, its work function equals ..... .

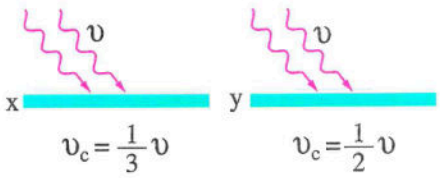
(a) 4.42×10^{-19} J (b) 2.21×10^{-19} J (c) 3.3×10^{-19} J (d) 6.625×10^{-19} J

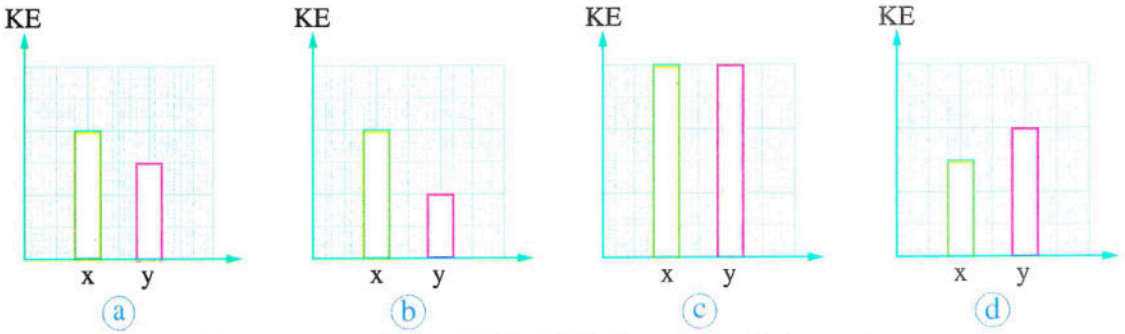
78 A light ray of wavelength 3000 Å falls on a metal surface and photoelectrons of maximum kinetic energy 0.5 eV are emitted. If another light ray of wavelength 2000 Å has fallen on the same metal surface, the maximum kinetic energy of the emitted photoelectrons becomes ..... .

(a) zero (b) less than 0.5 eV and greater than zero

(c) 0.5 eV (d) greater than 0.5 eV

79 The opposite figure shows a light of the same frequency and intensity has fallen on two different surfaces (x and y) of critical frequencies $\frac{1}{3}\upsilon$ and $\frac{1}{2}\upsilon$ respectively. Which of the following graphs represents the ratios of the kinetic energies of the electrons emitted from the two surfaces?





80 ✳ A light of frequency 6×10^{14} Hz has fallen on a metallic surface, so the maximum kinetic energy of the emitted electrons was 0.18 eV and when a light of frequency 1.6×10^{15} Hz has fallen on the same metal surface, the maximum kinetic energy for the emitted electrons was 4.32 eV, then Planck's constant is ..... .

(a) 6.625×10^{-34} J.s (b) 6.602×10^{-34} J.s (c) 6.624×10^{-34} J.s (d) 6.614×10^{-34} J.s

81 If the energy required to free an electron from a metallic surface is 3.968×10^{-19} J and when three monochromatic light beams of wavelengths 7000 Å, 6200 Å and 5000 Å respectively have fallen on that surface, then:

(i) The light beam which leads to the liberation of electrons from that surface has a wavelength of ..... .

(a) 7000 Å (b) 6200 Å (c) 5000 Å (d) all of them

(ii) The energy of the liberated electron equals ..... .

(a) 1.14×10^{-21} J (b) 3.97×10^{-21} J (c) 7.94×10^{-22} J (d) 7×10^{-22} J

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82 ✳ In two different experiments for studying the photoelectric phenomenon, two beams of electromagnetic radiation of frequencies 4×10^{15} Hz and 6×10^{15} Hz have fallen on two surfaces of the same metal, so the ratio of the maximum kinetic energy for the emitted electrons in the first experiment to that emitted in the second experiment was 1 : 3, then the critical frequency for that surface is ..... .

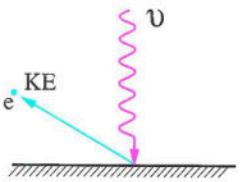
(a) 2×10^{15} Hz (b) 10^{15} Hz (c) 3×10^{15} Hz (d) 4×10^{15} Hz

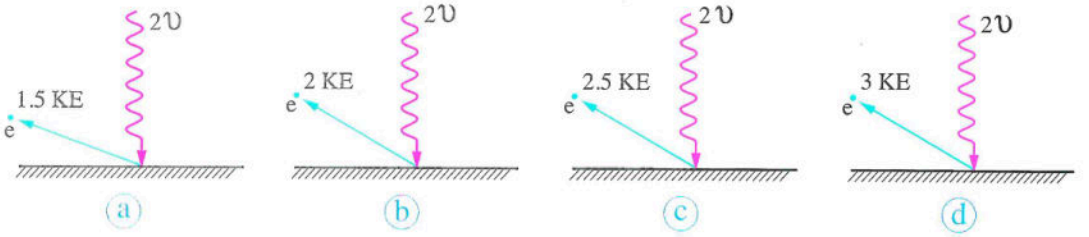
83 ✳ The opposite table shows the intensity of radiation and the frequency of three spectra (A, B and C) each one is used to illuminate a metallic surface of work function 3.056×10^{-19} J, so which of these spectra (A, B or C) can liberate maximum number of electrons in one second?

Spectrum	Frequency (Hz)	Intensity
A	3.5×10^{14} Hz	High
B	5.5×10^{14} Hz	Medium
C	7.5×10^{14} Hz	Weak

(a) A (b) B (c) C (d) B and C together.

84 The opposite figure shows the occurrence of the photoelectric phenomenon at a certain surface of a metal of work function $\frac{1}{2}$ hυ, which of the following figures expresses this surface?





(a) (b) (c) (d)

85 ✳ A light has fallen on a metallic surface of work function 3 eV, then:

(i) The minimum frequency of light that leads to the emission of photoelectrons is ..... .

(a) 4.5×10^{15} Hz (b) 1.21×10^{15} Hz (c) 7.25×10^{14} Hz (d) 4.5×10^{14} Hz

(ii) The longest wavelength of light that leads to the emission of photoelectrons is ..... .

(a) 1.07×10^{-7} m (b) 4.14×10^{-7} m (c) 6.67×10^{-8} m (d) 1.07×10^{-8} m

(iii) The frequency of light that leads to the photoelectrons emission of maximum kinetic energy 2 eV is ..... .

(a) 1.5×10^{14} Hz (b) 3.03×10^{14} Hz (c) 1.21×10^{15} Hz (d) 2.71×10^{15} Hz

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86 * The opposite table shows wavelengths for some spectral colors. When these colors fall on the cathode surface of a photoelectric cell of work function 2.2 eV, then:

Color	Wavelength (Å)
Red	6500
Yellow	5800
Green	5625
Blue	4500
Violet	4000

(i) The colors which cause the emission of photoelectrons when they fall on the cathode of the cell are ..... .

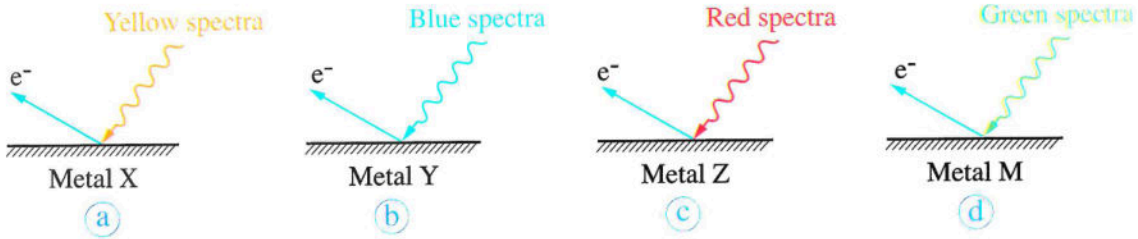
(a) red and yellow only (b) blue and violet only

(c) violet only (d) green, blue and violet

(ii) The maximum speed of the emitted photoelectrons when these colors fall on the cathode surface is ..... .

(a) 445×10^3 m/s (b) 13.38×10^3 m/s (c) 5.64×10^5 m/s (d) 5.41×10^4 m/s

87 In the following figures, electrons of the same maximum kinetic energy are emitted from the surface of each metal, which of the following metals has the greatest surface work function?



(a) (b) (c) (d)

88 * When a monochromatic light of wavelength (λ) has fallen on a metallic surface, the kinetic energy of the emitted electrons was 1.6×10^{-19} J and when another monochromatic light with wavelength $\frac{\lambda}{2}$ has fallen on the same surface, the kinetic energy of the emitted electrons was 6.4×10^{-19} J, then the work function of that surface equals ..... .

(a) 1.2×10^{-19} J (b) 1.6×10^{-19} J (c) 2.4×10^{-19} J (d) 3.2×10^{-19} J

89 * When the frequency of the incident photons which fall on the surface of a photoelectric cell increases by 20%, the kinetic energy of the electrons which are emitted from the metal surface increases from 0.5 eV to 0.8 eV, so the work function of that metal is ..... .

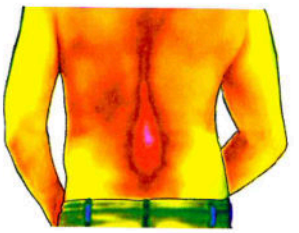
(a) 6.4×10^{-19} J (b) 4.8×10^{-19} J (c) 3.2×10^{-19} J (d) 1.6×10^{-19} J

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Second Essay questions

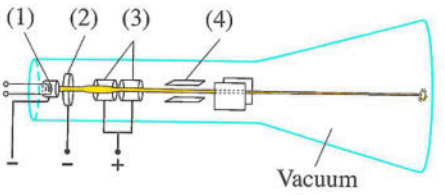
Blackbody radiation

1. **Compare between:** The radiation emitted from the Sun (luminous body) and the radiation emitted from the Earth (non-luminous body) (in terms of the region of the wavelength which has the maximum intensity of radiation).
2. **How did** Planck explain the low intensity of the radiation emitted by an incandescent body at high frequencies?
3. **Give reason:** A person can see the radiation emitted from a glowing coal in a dark room, while some components of the room cannot be seen.
4. While heating up an iron rod gradually, it glows producing a radiation whose color varies with temperature, **what is** the dominant color of radiation that appears first on the rod during heating it up? **And what is the explanation** for that?
5. **Mention one application (or one use) for:**
   (a) Infrared rays.
   (b) The microwaves.
   (c) The thermal radiation emitted from the human body.
6. The opposite figure shows an image taken by a special type of camera for a human body. **Explain** the meaning of the variation of color of each part from the other. **What is the scientific idea** for that kind of photography?



Thermionic emission and photoelectric effect

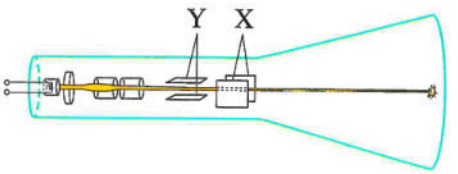
7. **The opposite diagram represents the structure of a cathode ray tube:**



   (a) **What** is the component from which the electrons are emitted? **And what** is the condition for their emission?
   (b) **How does** component (2) control the brightness of the fluorescent screen at each given point?
   (c) **Mention two functions** executed by component (3) on the electrons ray.
   (d) **What** is the factor that controls the speed of electrons at reaching component (3)?
   (e) **Why** is the tube evacuated of air?

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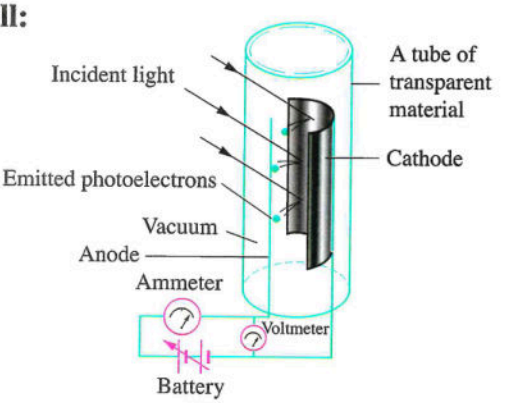
8 The opposite figure represents a sketch for a cathode ray tube working perfectly, **what happens to the cathode rays inside the tube, when:**



(a) The plates X stop working?

(b) The plates Y stop working?

9 **The opposite figure represents a photoelectric cell:**



(a) Explain why the anode is made of a thin wire.

(b) What property of the incident light can be increased to increase the ammeter reading?

(c) What are the factors that affect each of:

① The work function of the surface of the metal?

② The maximum kinetic energy of the electrons emitted from the cathode?

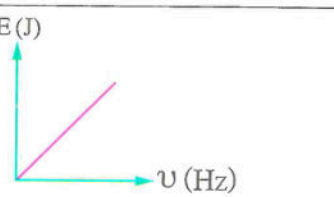
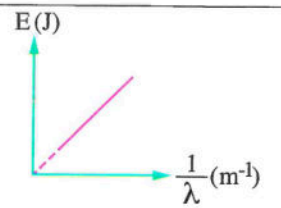
10 **Explain:** Electrons may not be emitted from the surface of a metal while being exposed to a certain electromagnetic radiation for a long-time interval.

11 **Explain why** when a very bright red light falls on the surface of cesium no photoelectrons are emitted, while when a faint blue light falls photoelectrons are emitted from the surface.

12 **Compare between:** The cathode ray tube and the photoelectric cell (in terms of the type of energy that causes liberation of electrons from the cathode).

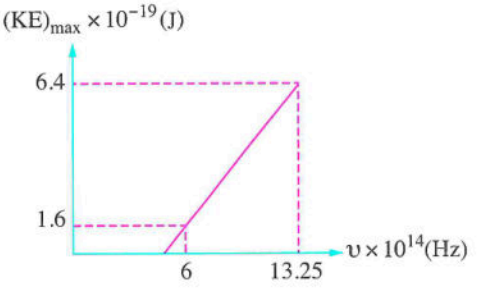
13 If the work function of copper is 4.4 eV, **can** visible light liberate electrons if it falls on a copper surface? **Explain your answer.**

14 **Write the mathematical relation and the equivalent to the slope for each of the following graphs:**

(a) 	**(b)** 
(Where: (E) is photon energy, $\frac{1}{\lambda}$ is the reciprocal of wavelength of photon, (υ) is the frequency).	

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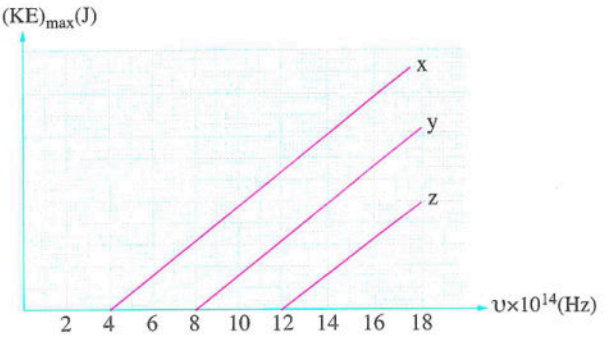
15 A scientist has represented the values that he obtained from an experiment to study the photoelectric effect of a metal as shown in the opposite graph, **find:**



(a) Planck's constant.

(b) The critical frequency of the metal.

16 The following graph represents the relation between the maximum kinetic energy of the emitted electrons from the surface of three metals (x, y and z) and the frequency of the incident light on each of them, depending on the graph:



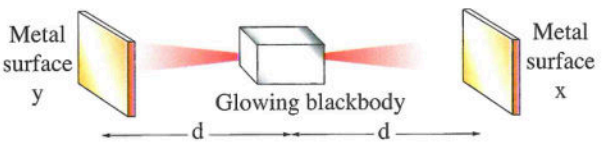
(a) **Calculate** the work function of the metal y.

(b) **What** is the minimum frequency of an electromagnetic radiation required to liberate an electron from any of these metals?

(c) If a light ray of a certain frequency has fallen and liberated electrons from the surface of each of the three metals, **which of these metals** liberates electrons that have the highest kinetic energy? **Explain your answer.**

17 A light of frequency υ_1 has fallen on the surface of a metal x of work function $(E_w)_x$, so electrons of maximum kinetic energy $(KE)_1$ are emitted. Another light of frequency υ_2 has fallen on the surface of a metal y of $(E_w)_y$, so electrons of maximum kinetic energy $(KE)_2$ are emitted. If $(E_w)_y < (E_w)_x$, $(KE)_2 = (KE)_1$, **then which** of the two lights has higher frequency? **Explain your answer.**

18 The opposite figure represents a glowing blackbody placed in the middle of the distance between two metal surfaces (x and y) and the work function of the metal (x) was double the work function of the metal (y), so electrons were emitted from the two surfaces, **from which** surface are electrons emitted at a greater rate? **And why?**

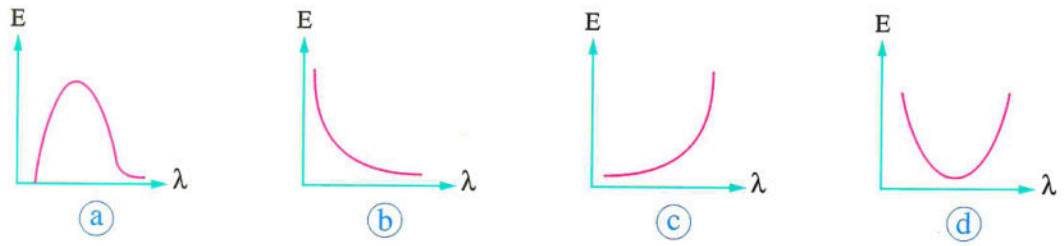


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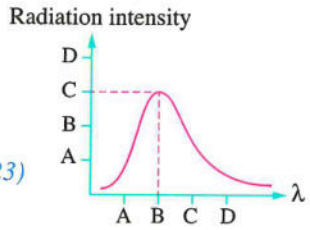
Questions of the Previous Years Exams on Lesson 1

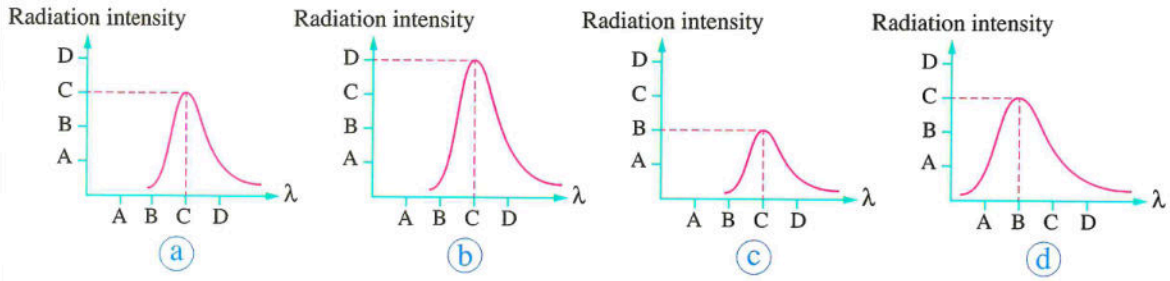
Blackbody radiation

1 Which of the following graphs represents the relation between the energy of a photon (E) in the radiation emitted from a blackbody and its wavelength (λ)? *(1st session 23)*



2 The figure illustrates Planck's distribution curve for a hot body. If the body is left to cool down, the curve changes to figure ..... .
(Knowing that: the figures do not have the same scale) *(2nd session 23)*





3 The opposite graph represents Planck's distribution for the radiation of a hot body at 6000 K where the wavelength associated with the maximum radiation is λ_o, so the wavelength associated with the maximum radiation becomes 2 λ_o when the temperature ..... .

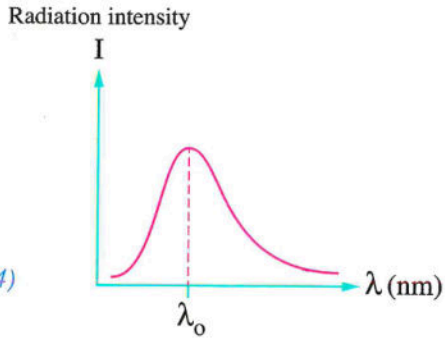
(2nd session 24)

(a) decreases by 1500 K

(b) increases by 3000 K

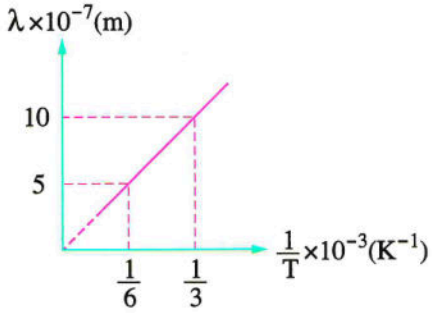
(c) decreases by 3000 K

(d) increases by 1500 K



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4 The opposite figure shows the relation between the wavelengths (λ) for maximum intensity of radiation and the reciprocal of the temperature ($\frac{1}{T}$) on Kelvin scale, so the wavelength for maximum intensity of radiation at 2000 K is ..... . *(1st session 24)*

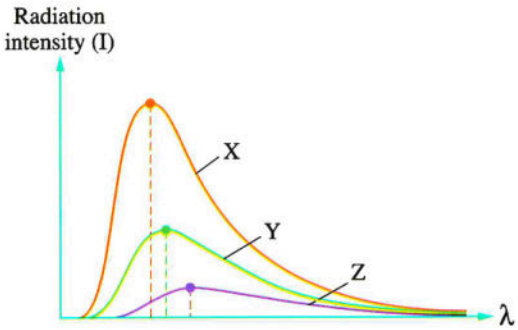


(a) 15000 Å
(b) 20000 Å
(c) 15000 nm
(d) 20000 nm

5 A laser with a wavelength of 200 nm is emitted. How many laser photons are emitted if their total energy is 1 joule ? *(Practice 25)*

(a) 8.32×10^{18} (b) 1.01×10^{18} (c) 1.36×10^{18} (d) 9.95×10^{19}

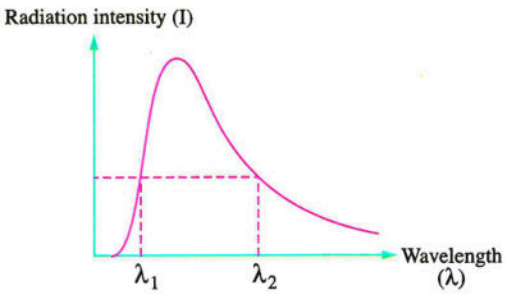
6 The graph represents the relationship between the intensity of electromagnetic radiation (I) and the wavelength (λ) of the radiation emitted from radiation emitted from three glowing objects, X, Y, and Z. *(1st session 25)*



Which of the following statements is true?

(a) Object X has the highest temperature of the three objects.
(b) Object Y has the highest temperature of the three objects.
(c) Object Z has the highest temperature of the three objects.
(d) The three objects have the same temperature.

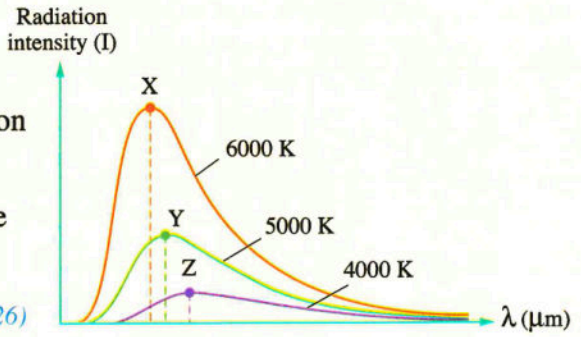
7 The opposite graph represents Planck's distribution of solar radiation, so the radiation having wavelength λ_2 may be of ..... rays, and the energy of one of its photons is ..... the energy of a photon of the radiation having wavelength λ_1. *(2nd session 25)*



(a) ultraviolet, equal to
(b) ultraviolet, greater than
(c) infrared, less than
(d) infrared, equal to

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8 The opposite graph represents the relation between the intensity of electromagnetic radiation (I) and the wavelength (λ) of radiation emitted from three glowing bodies. The ratio between the wavelengths corresponding to the maximum radiation intensity ($\lambda_X : \lambda_Y : \lambda_Z$) is ..... *(Practice 26)*



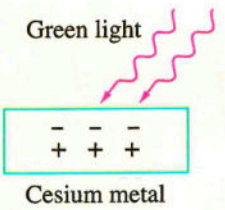
(a) 6 : 5 : 4 (b) 4 : 5 : 6
(c) 12 : 10 : 5 (d) 10 : 12 : 15

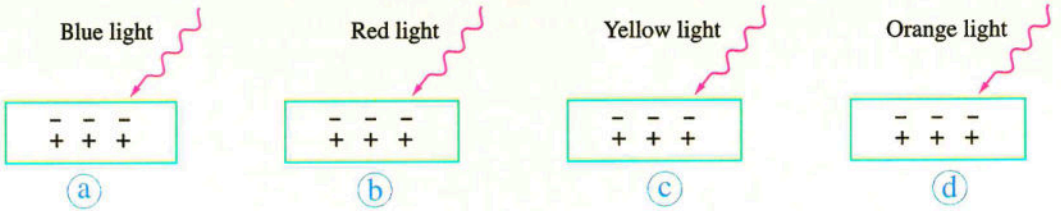
9 Two glowing bodies (x), (y) are made of the same material and having the same surface area with temperatures of 2000 K and 6000 K, respectively, produce electromagnetic radiation. If the frequencies at maximum radiation intensity of each of them are υ_x and υ_y, respectively, then the ratio of $\frac{\upsilon_x}{\upsilon_y}$ equals ..... *(Practice 26)*

(a) $\frac{1}{3}$ (b) $\frac{3}{1}$ (c) $\frac{3}{4}$ (d) $\frac{4}{3}$

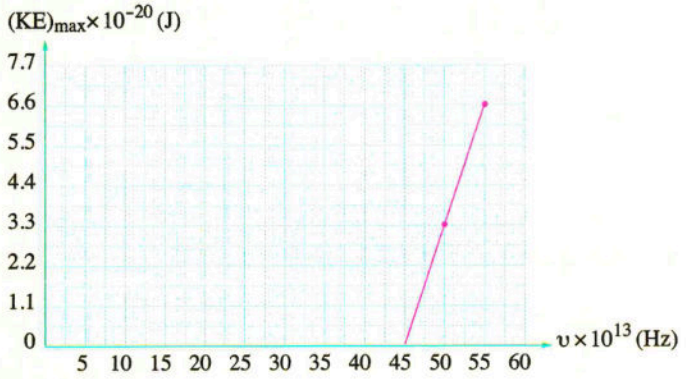
Thermionic emission and photoelectric effect

10 The opposite figure represents a green light falling on the surface of cesium metal, where electrons are barley liberated with zero KE, so in which figure from the following electrons will be liberated gaining KE? *(Exp. June 21)*





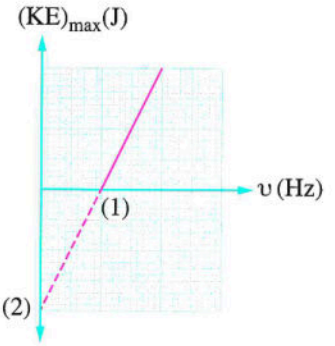
11 The opposite graph represents the relation between the maximum kinetic energy of the electrons emitted from a photoelectric cell and the frequency of the incident light on the cathode, so which one of the following wavelengths causes emission of electrons with kinetic energy = 6.6×10^{-20} J? *(Exp. June 21)*



(a) 5.5×10^{-7} m (b) 5.3×10^{-7} m (c) 5.6×10^{-7} m (d) 5.7×10^{-7} m

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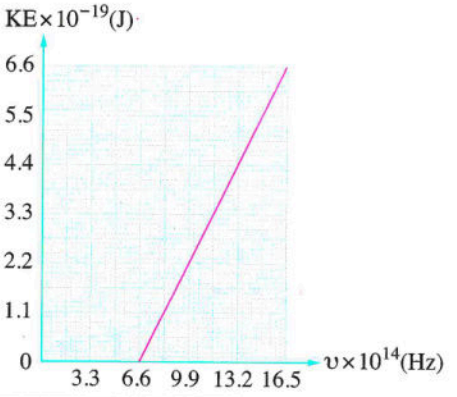
12 The opposite graph represents the relation between the maximum KE for the emitted electrons from a metal surface and the frequency of the incident light, so the measuring unit of the quotient of $\frac{\text{the value of point (2)}}{\text{the value of point (1)}}$ is ..... . *(1st session 21)*



(a) $kg.m^2.s$ (b) J/s

(c) $kg.m^2.s^{-1}$ (d) $kg.m.s^{-1}$

13 The opposite graph represents the relation between the maximum kinetic energy of the electrons emitted from a photoelectric cell and the frequency of the incident photons on the cathode, so the work function of the surface equals ..... . *(1st session 21)*



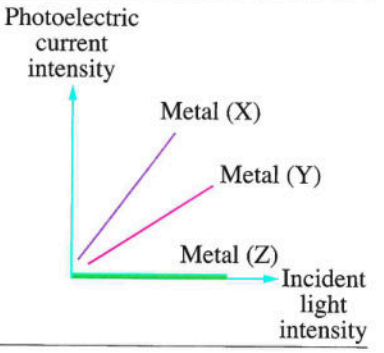
(a) 2.7 eV

(b) 0.27 eV

(c) 0.027 eV

(d) 27 eV

14 Three photoelectric cells of different cathode materials (X, Y and Z), the graph illustrates the relation between the photoelectric current intensity and light intensity falling on each cell, which metal has a critical frequency greater than the frequency of the incident light? *(2nd session 21)*

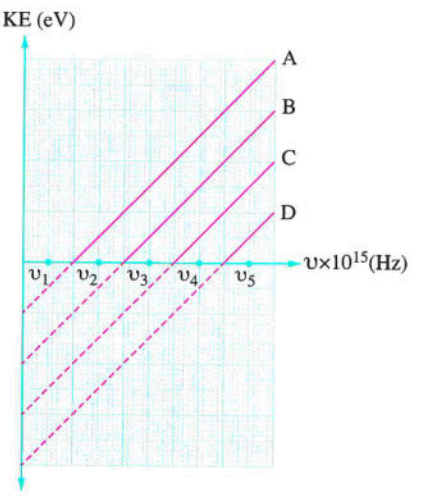


(a) Metal (X). (b) Metal (Y).

(c) Metal (Z). (d) All metals.

15 The graph represents the relation between maximum kinetic energy for the emitted electrons from the surfaces of 4 different metals (A, B, C and D) and the frequency of the light falling on them.

Which frequency causes electrons to get emitted from metals (A and B), but does not cause electron emission from metals (C and D)?

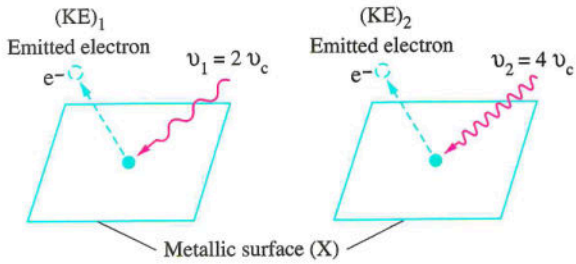


(a) υ_3 (b) υ_5

(c) υ_2 (d) υ_4 *(1st session 22)*

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16 The figure shows a metallic surface (X), the critical frequency of its metal is (υ_c). Once a photon of frequency ($\upsilon_1 = 2\,\upsilon_c$) falls on the surface, a photoelectron is emitted with maximum kinetic energy (KE_1). When the previous photon is replaced by another one of frequency ($\upsilon_2 = 4\,\upsilon_c$), a photoelectron is emitted with maximum kinetic energy (KE_2). Then, the ratio of $\left(\frac{KE_1}{KE_2}\right)$ = ...



(1st session 22)

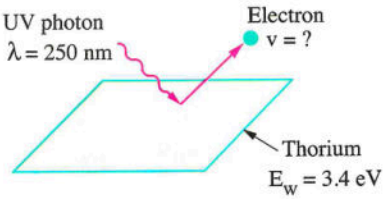
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$

17 A metallic surface, the work function of its metal is (E_w). Once a photon of energy (E_1) that equals three times as the metal work function falls on the surface a photoelectron is emitted with maximum speed (v). When the previous photon is replaced by another one of energy (E_2) that equals seven times as the metal work function, the photoelectron will be emitted with maximum speed of ..... .

(2nd session 22)

(a) $\sqrt{3}$ v (b) 3 v (c) $\sqrt{6}$ v (d) 6 v

18 Using the data illustrated in the figure, the maximum speed of the emitted photoelectron due to the falling of UV photon on the surface of thorium metal = ...



(2nd session 22)

(Given that: $m_e = 9.1 \times 10^{-31}$ kg , $e = 1.6 \times 10^{-19}$ C, $h = 6.625 \times 10^{-34}$ J.s, $c = 3 \times 10^8$ m/s)

(a) 7.43×10^4 m/s (b) 7.43×10^6 m/s (c) 7.43×10^5 m/s (d) 7.43×10^3 m/s

19 Electromagnetic radiation was incident on three surfaces made of different metals A , B and C, as a result, photoelectrons were emitted, if the work function $E_{w(C)} > E_{w(B)} > E_{w(A)}$. Which of the following choices represents the correct arrangement of the maximum kinetic energy for the photoelectrons? *(Exp. 23)*

(a) $KE_{(B)} < KE_{(A)} < KE_{(C)}$ (b) $KE_{(C)} < KE_{(B)} < KE_{(A)}$

(c) $KE_{(A)} < KE_{(C)} < KE_{(B)}$ (d) $KE_{(C)} < KE_{(A)} < KE_{(B)}$

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20 When the potential of the grid in the cathode ray tube changes from (– 4 V) to (– 12 V) with the constancy of the potential difference between the anode and the cathode, which of the following choices is correct? *(1st session 24)*

	The number of electrons that pass through the grid	The brightness of the fluorescent screen
(a)	Decreases	Increases
(b)	Increases	Increases
(c)	Decreases	Decreases
(d)	Increases	Decreases

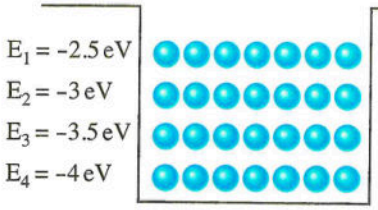
21 If the photons of a light beam fall on a cathode of photoelectric cell with a rate (ϕ_L) and frequency (υ), the intensity of the produced photocurrent is 3 mA. So, by increasing the rate of the incident photons of the same light source to double, the electric current and the work function might be ..... . *(1st session 24)*

	The intensity of photocurrent	Work function
(a)	3 mA	remains constant
(b)	3 mA	decreases to half
(c)	6 mA	remains constant
(d)	9 mA	increases to double

22 If the potential difference between the anode and cathode in cathode ray tube is 500 V, the maximum kinetic energy for the emitted electrons from cathode equals ..... . *(Practice 25)*

(a) 4×10^{-17} J (b) 8×10^{-17} J (c) 12×10^{-17} J (d) 16×10^{-17} J

23 The opposite figure represents a diagram of binding energy of electrons in a metallic surface, so the work function of the surface equals ..... . *(Practice 25)*



(a) 2.5 eV (b) 3 eV (c) 3.5 eV (d) 4 eV

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24 When a light falls on metallic surface and electrons are emitted. **What will happen for** the work function and the kinetic energy of the emitted electrons if we use a light of higher frequency on the same metal surface? *(Exp. 23)*

.....

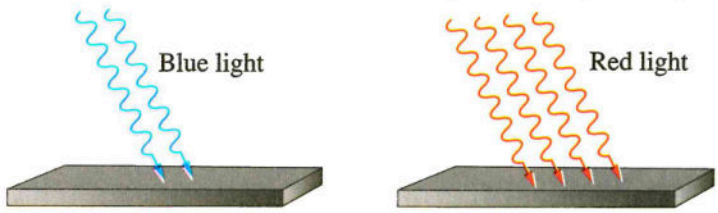
.....

25 A monochromatic light of frequency 6×10^{14} Hz falls on the cathode of a photoelectric cell, so electrons of maximum kinetic energy (1 eV) are emitted. When another light of frequency (X) Hz falls on the same cathode of the photoelectric cell, the maximum kinetic energy of the emitted electrons is (0.38 eV), **calculate** the light frequency (X). *(1st session 23)*

.....

.....

26 The figure illustrates photoelectric effect in two cases (X and Y). In case (X), dim blue light falls on the metal surface, and in case (Y), high-intensity red light falls on the same metal surface. *(1st session 25)*



Case (X) **Case (Y)**

Assuming that electrons are emitted from the metal surface in both cases, compare the photoelectrons emitted in the two cases in terms of:

(1) Maximum speed of their emission. **(2)** Rate of electron emission

.....

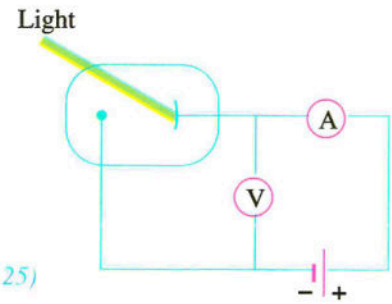
.....

27 The figure shows a circuit of a photoelectric cell on which a beam of a monochromatic light is incident. The ammeter reads a weak photoelectric current in the circuit.

To increase the photoelectric current in the circuit, **what** action should be taken regarding:

(1) The way of connecting the battery to the circuit?

(2) The characteristics of light falling on the cell? *(2nd session 25)*



.....

.....

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